

Skyla Trousdale  
CART 498  
Dr Gabriel Vigliensoni

#### Reflection for Assignment #4

I noticed that the model breaks when the  $n$  is negative, the  $n$  is a decimal, or both. I suspect this is because I am using a small model where the numbers are seen as tokens / oversimplified and broken down like the simple predictive models we were using in class. While small numbers are often not made into large tokens, large numbers are, same with numbers with decimals and negatives. This is likely what is happening with the small models.

I also suspect the model may mark itself as wrong sometimes when the answer is correct, because again it is weighing large numbers that it is interpreting as a sort of text/tokens. Due to the self-deprecating command of my prompt, any errors are immediately flagged which turns up the temperature and the top\_p even if there was no error. This puts another layer of dependency on the ability of the small language model AI to judge itself and make calculations.

However, the model is largely successful for numbers that are small and simple. This includes small integer, non-negative input for  $n$ , and small iterations for  $i$ . With less room for mistakes granted by the small  $i$  value, the temperature and top\_p both stayed low which likely increased the success of the next calculation.